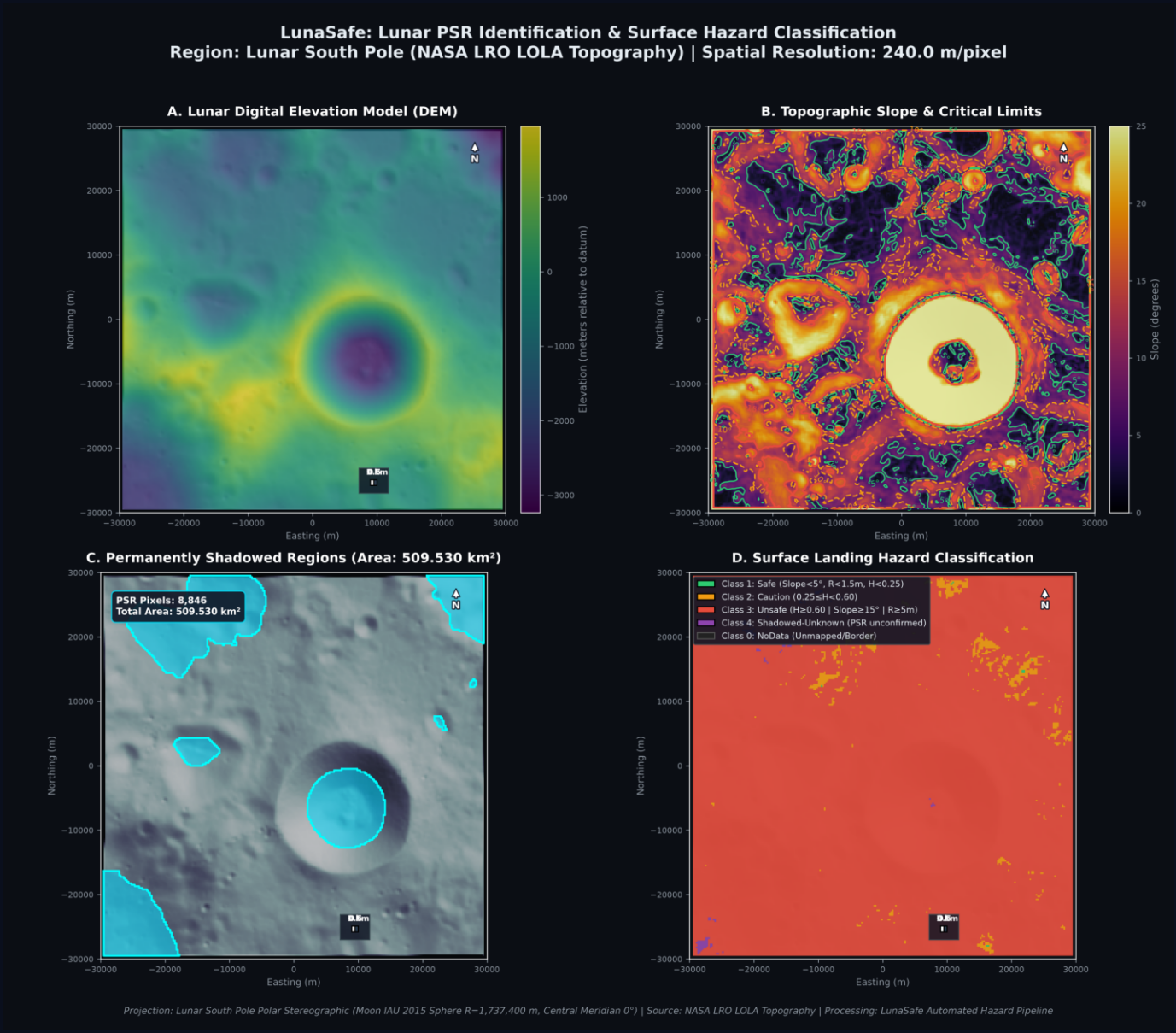
LUNASAFE MISSION CONTROL

Autonomous Lunar Landing & Hazard Classification Mission Report

Dataset Provenance: [REAL NASA LOLA] Real NASA LRO LOLA DEM (South Pole, 240 m/px)

CRS: Moon South Pole Polar Stereographic (IAU 2015 Sphere R=1,737,400m) | Generated: 2026-10-21 21:47:49 UTC



EXECUTIVE SAFETY METRICS SUMMARY

- Safe Zone (Class 1): **0.63 km² (0.0% of valid terrain)**
- Caution Transition (Class 2): **64.92 km² (1.9%)**
- Hard Unsafe Overrides (Class 3): **3411.30 km² (97.9%)**
- Shadowed Cold Trap (Class 4): **8.87 km² (0.2%)**

9-Point Rule Audit:

PASSED (0 VIOLATIONS)

Obstacle Clearance:

R_safe > 100m Envelopes

LUNASAFE MISSION CONTROL

Comprehensive Surface Telemetry & Mathematical Verification Audit

Verification against strict lunar flight certification constraints (IAU 2015 Lunar South Pole Spheroid)

1. SURFACE CLASSIFICATION QUANTITATIVE BREAKDOWN

Hazard Class	Pixels	Area (km²)	% Valid Terrain	% Total Raster
Class 1: Safe	11	0.63 km²	0.0%	0.0%
Class 2: Caution	1,127	64.92 km²	1.9%	1.8%
Class 3: Unsafe	59,224	3411.30 km²	97.9%	94.8%
Class 4: Shadowed-Unkn	154	8.87 km²	0.2%	0.2%
Class 0: NoData	1,984	114.28 km²	0.0%	3.2%

2. 9-POINT DETERMINISTIC RULE CONSISTENCY AUDIT

Rule ID	Mathematical Safety Criterion	Violations	Audit Status
Rule 1	Safe pixel slope must strictly b	0	PASSED
Rule 2	Safe pixel roughness must stri	0	PASSED
Rule 3	Safe pixel multi-criteria screen	0	PASSED
Rule 4	No Safe pixel may be located v	0	PASSED
Rule 5	Hard Slope Override: Any pixel	0	PASSED
Rule 6	Hard Roughness Override: Any	0	PASSED
Rule 7	Hazard Score Override: Any pi	0	PASSED
Rule 8	Permanently Shadowed terrain	0	PASSED
Rule 9	NoData missing elevation pixe	0	PASSED

3. SCIENTIFIC ASSUMPTIONS, METHODOLOGY & OPERATIONAL LIMITATIONS

- Topographic Gradient: Horn (1981) 8-neighbor central finite-difference operator applied to DEM altimetry.
- Surface Micro-Relief: Standard deviation of local elevation residuals represents micro-relief within lander footpads.
- Multi-Criteria Screening Index: Live formula H = Ws·(Slope/15°) + Wr·(Rough/5m) + Wd·ShadowFraction with weights Σ=1.0.
- Spatial Resolution Limitation: DEM cell resolution (240 m/px real or 20 m/px synthetic) does not resolve boulders < 1 meter.
- Illumination Physics: Single-frame elevation lighting is an instantaneous preview. Long-term cold-traps require full diurnal tracking.
- Geotechnical Bearing Capacity: Regolith bulk density and shear strength are unmodeled; surface firmness requires tactile radar/lidar.
- Flight Clearance Rule: Candidate landing sites require 100% compliance over 100 m radius ellipse with > 40 m standoff from shadows.